

MATH 425 Differential Geometry

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Outline

- 1 Introduction to Algebraic Geometry
 - Overview

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Algebraic sets

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$$\begin{cases} p_1(x_1, \dots, x_n) = 0 \\ p_2(x_1, \dots, x_n) = 0 \\ \dots \\ p_m(x_1, \dots, x_n) = 0 \end{cases}$$

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Notation:

$$V(S) = \{(a_1, \dots, a_n) \in \mathbb{C}^n : p(a_1, \dots, a_n) = 0 \ \forall p \in S\}.$$

Example: conic

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- Principal branch:

$$g_I(z) = \sqrt{z} = r^{1/2} e^{i\theta}, \quad r \geq 0, \theta \in (-\pi, \pi)$$

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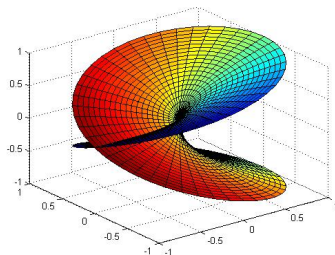
$$V_I = \{(x, g_I(x)) : x \in B_I\} \quad \text{and} \quad V_{II} = \{(x, g_{II}(x)) : x \in B_{II}\}$$

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- Region II:

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$$U_I = \{(a, f_I(a)) : a \in D_I\} \quad \text{and} \quad U_{II} = \{(a, f_{II}(a)) : a \in D_{II}\}$$

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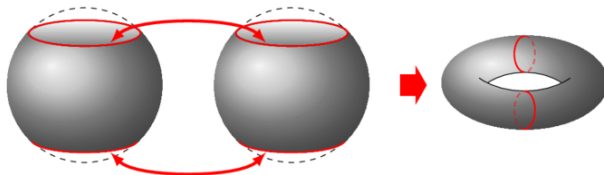
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Creates a torus

practically k holes on the torus we get $\chi = -k$. If we want to pass to a surface of genus 2, we l



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$V(I(A)) =$ Zariski closure of A

Points and maximal ideals

$$S = \{p_1(x_1, \dots, x_n), \dots, p_m(x_1, \dots, x_n)\}$$

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$$\langle x - a \rangle = \{(x - a)q(x) : q \in \mathbb{C}[x]\}$$

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- the maximal ideals of R are

$$\langle x-a, y-b \rangle = \{(x-a)q_1(x, y) + (y-b)q_2(x, y) : q_1, q_2 \in \mathbb{C}[x]\}$$

for each $a \in \mathbb{C}$

- to contain xy , must have $a = 0$ or $b = 0$

Algebraic and geometric dimension

The algebraic dimension or **Krull dimension** of a commutative ring R the largest integer d for which there are prime ideals

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Theorem: The Krull dimension of R/I is the complex dimension (half the real dimension) of $V(S)$.